

MD. ESHA RUHULLA

GEOTECHNICAL ENGINEERING | NUMERICAL MODELLING

CONTACT

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EDUCATION

Islamic University of Technology
B.Sc. in Civil & Environmental Engineering
CGPA 3.37 / 4.00
2022 – 2026

Rangpur Govt. College
Higher Secondary Certificate (Science)
GPA 5.00 / 5.00
2019 – 2021

Police Lines School & College, Rangpur
Secondary School Certificate (Science)
GPA 5.00 / 5.00
2017 – 2019

TECHNICAL SKILLS

- **Numerical Modelling:** PLAXIS 3D, PLAXIS 2D, ETABS
- **Scientific Computing:** Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB
- **Machine Learning:** DeepXDE, PyTorch (physics-informed neural networks)
- **Computer Aided Design:** AutoCAD
- **GIS:** ArcGIS
- **Methods:** Complex variable elasticity, conformal mapping, settlement trough & ground loss analysis
- **Documentation:** $\LaTeX$, MS Excel, Word, PowerPoint

EXTRA CURRICULAR ACTIVITIES

- **Vice President**—CENNOVATION 2026, Dept. of CEE, IUT
- **Head, PR & Communication**—ACI IUT Student Chapter, 2025–2026

CERTIFICATIONS & TRAININGS

- Industrial Training, Geotechnical & Consultancy Practice — DOHWA Engineering (2025)

REFERENCES

- **Prof. Dr. Hossain Md. Shahin**—Professor & Thesis Supervisor, CEE, IUT
- **Dr. Md. Imran Kabir**—Associate Professor, CEE, IUT
- **Mohammad Abu Umama**—Lecturer, CEE, IUT
- *Contact details on request.*

CAREER OBJECTIVE

Final-year Civil and Environmental Engineering student at IUT working in geotechnical engineering, with a thesis on forward prediction of tunnelling-induced ground settlement by analytical elasticity and physics-informed machine learning, and design experience in three-dimensional finite element modelling of foundations. Comfortable moving between closed-form mechanics, numerical modelling and field practice, and seeking research or engineering work in underground construction, soil–structure interaction and computational geomechanics.

UNDERGRADUATE THESIS

Forward Prediction of Tunnelling-Induced Settlement 2025 – 2026
A unified displacement function and a physics-informed neural network validated against controlled model tests — Supervisor: Prof. Dr. Hossain Md. Shahin

- Tested whether settlement above a shallow tunnel can be predicted forward, from the tunnel-wall deformation, rather than by back-fitting a measured trough — the assumption behind the empirical Gaussian curve used in practice.
- Implemented the unified displacement function as a complex-variable solution of the elastic half-plane: conformal mapping onto an annulus, Laurent-series potentials and least-squares collocation. Maximum boundary residual 10^{-14} ; published field case reproduced to within 0.02 mm.
- Rebuilt the physics-informed neural network in DeepXDE as mixed first-order elasticity, identifying non-dimensionalisation and logarithmically graded collocation as the two measures needed for convergence. Matched the analytical benchmark to 0.77% of peak settlement.
- Validated against model tests where the tunnel-wall movement was imposed and therefore known, giving an out-of-sample score: peak settlement within 12% for uniform convergence, and an empirical width correction raising mean R^2 from 0.41 to 0.91 across six cases.

PUBLICATIONS

- A. Mahmud and M. E. Ruhulla, “Forward prediction of tunnelling-induced settlement: a unified displacement function and a physics-informed neural network validated against controlled model tests,” *International Conference on Civil Engineering (ICCE 2026)*, Dhaka, 17–19 December 2026. **Submitted, under review.**

DESIGN & MODELLING PROJECTS

Final Year Design Project (FYDP) — Group 6 2026
Foundation design of a multi-storeyed building — PLAXIS 3D

- Built three-dimensional finite element models of a pile foundation and a raft foundation on a common soil profile and loaded each to failure through staged construction.
- Pile foundation: ultimate bearing capacity 1879.85 kN at 21.9 mm maximum total displacement. Raft foundation: 2517.30 kN at 117.8 mm.
- Compared the two systems on capacity against deformation and recommended design parameters on settlement and bearing-capacity criteria.

Analysis Code for Tunnelling Settlement Prediction 2025 – 2026
Complex-variable solver and PINN implementation — Python

- Reproducible analysis pipeline in NumPy/SciPy and DeepXDE; every measured value read from source data rather than digitised from published figures.

INTERNSHIPS

Industrial Trainee — DOHWA Engineering Co., Ltd. Oct 2025
Geotechnical & consultancy practice — Bangladesh Branch, Dhaka

- Four-week training across planning, design, project management and procurement workflows at an international consultancy.
- Assisted in preparing EOIs and RFPs and in proposal evaluation under Bangladesh’s Public Procurement Rules (PPR); practised quantity surveying and cost estimation.
- Studied geotechnical investigation practice for foundation design; site visit to the Pagla Sewage Treatment Plant, Narayanganj, under the Dhaka Sanitation Improvement Project.